



SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **Chakradhar Peela 192412400** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Strategic Framework: Analyzing Goal-Balancing Competencies in Complex Systems

1. Theoretical Foundation: The Multidimensional Nature of Complex Problem Solving (CPS)

In the contemporary "information explosion" era, strategic proficiency is no longer measured by static reasoning but by the capacity to navigate Complex Problem Solving (CPS) environments. High-proficiency individuals are differentiated by their ability to maintain performance within dynamic systems where rules are not provided *a priori* but must be induced through active exploration. As task environments scale in complexity, the solver's capacity to integrate information reduction with model formation becomes the primary determinant of success. The following table synthesizes the traditional Five-Dimensional Model against the PISA 2012 framework, highlighting the critical gaps in current assessment architectures.

Dimension Category	Five-Dimensional Model (Funke)	PISA 2012 Model (OECD)
Information Gathering	Active interaction to identify latent rules.	Active interaction to identify latent rules.
Exploring and Understanding	Building initial mental models.	Building initial mental models.
Mental Modeling	Synthesizing complex feedback.	Synthesizing complex feedback.
Information Reduction & Model Formation	Creating internal representations.	Creating internal representations.
Execution	Manipulating variables to achieve states.	Manipulating variables to achieve states.
Control	Moving toward a goal state.	Moving toward a goal state.
Planning and Executing	Strategy adjustment.	Strategy adjustment.
Monitoring and Reflecting	Evaluating progress.	Evaluating progress.
Strategic Priority	The balancing of competing objectives.	The balancing of competing objectives.
Prioritization of Goals	(Missing link: Not explicitly represented as a unique dimension)	(Missing link: Not explicitly represented as a unique dimension)

Critiquing the "Clear-Goal" Limitation

Historically, CPS research has relied on simplified models like MicroDYN, which emphasize knowledge acquisition within linear, non-conflicting parameters. While effective for measuring basic rule induction, such models fail to elicit **Operative Intelligence** because they remove the "Conflict Layer" inherent in real-world systems. In MicroDYN, goals are typically independent and clear; however, true complex proficiency requires the ability to handle multiple, often contradictory, objectives. Justifying the inclusion of "Multiple Goals Balancing" (MGB) as a core competency allows psychometricians to interpret behavioral log data not merely as a record of accuracy, but as a sequence of strategic trade-offs and priority shifts.

2. Case Study Context: The PISA 2012 "Ticket" Task Architecture

The PISA 2012 "Ticket" task is a high-fidelity simulation of a complex vending system. The task architecture involves 24 ticket types across a train network (city vs. country), fare types (full vs. concession), and trip counts (daily vs. individual 1–5). This configuration creates a multi-layered goal environment.

The Tiered Goal Architecture (Subtask 3)

In the third subtask, solvers are confronted with three distinct goal layers:

- **The Demand Goal:** Successfully selecting a ticket that meets specific ride needs (city subway, two trips).
- **The Price Goal:** Seeking the lowest possible cost in "zeds" for the required travel.
- **The Availability Goal:** Discovering valid ticket types through trial and error, as certain combinations are systemically restricted.

The Conflict Layer

The strategic complexity of the Ticket task emerges from the direct conflict between **Price** and **Availability**. While concession tickets represent the optimal price point, they are unavailable for city subway travel. A solver who pursues price advantage blindly will reach a logical impasse. Success requires a "satisfactory trade-off" strategy, where the solver recognizes that Availability and Demand must be prioritized over the pursuit of a price discount. These mechanics provide the structured behavioral data required for deep-sequence analysis.

3. Methodology: Extracting Behavioral Indicators from System Logs

To ensure psychometric interpretability, this framework adopts a **theoretically-driven log data analysis**. By mapping behavioral sequences to specific psychological constructs rather than using "black box" data mining, we maintain the transparency required for educational evaluation.

Segmentation and Coding Process

Behavioral sequences are segmented into "plans," bounded by the solver clicking "Purchase" or "Cancel." Each plan is coded using the following indicators:

- **Demand Goal Pursuit:** Coded as **1** if the plan meets the specific ride requirements; **0** otherwise.
- **Price Goal Pursuit:** This is a relative metric dependent on the solver's unique **Knowledge Base**. A plan is coded as **1** if the solver "Cancels" after seeing the price (dissatisfaction) or if they "Purchase" a ticket that is the *cheapest known option* based on their entire history of previous trials (including failed attempts).
- **Availability Goal Pursuit:** This tracks strategic exploration rather than random clicks. Using a three-plan sequence (Adjustment 1–2 vs. Adjustment 2–3), if the solver targets different ticket attributes across these transitions (e.g., switching from fare type to trip count after a failure), it is coded as **1**, indicating a deliberate search for available states. The ratios of these pursuits against total plans serve as the primary structured variables, transforming raw process data into standardized behavioral metrics.

4. Statistical Taxonomy: Differentiating Solver Proficiencies

To identify behavioral clusters, we employ the **Partitioning Around Medoids (PAM)** algorithm. PAM is preferred over K-Means in psychometric contexts due to its robustness against the noise and outliers typical of human behavioral data.

Cluster Analysis and Goal Pursuit z-Scores

The analysis identified 10 distinct groups. The following table highlights the critical differences between "Successful" and "Poor" solvers.

Proficiency Level	Group Examples	Demand Goal
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(z) | Availability Goal (z) | Price Goal (z) | Task Score || ----- | ----- | ----- | ----- | ----- ||
Successful Solvers | Groups 1–2 | High Positive (+0.6) | High Positive (+0.6 to +1.3) | Variable
 | 0.88 || **Demand-Locked** | Group 10 | **Positive (+0.66)** | **Extreme Negative (-2.02)** | High
 Positive (+1.27) | **0.18** || **Ineffective Solvers** | Group 9 | High Negative (-3.05) | Positive (+0.52)
 | Positive (+0.43) | 0.24 |

Interaction Effects and the "Price Trap"

Regression analysis ($R^2 = 0.329$ for task score) reveals a clear hierarchy of effects:

- **The Hierarchy Effect:** The Demand Goal ($\beta = 1.041$) is the primary gatekeeper, significantly out-weighting the Availability Goal ($\beta = 0.702$). Success is predicated on first securing the Demand target.
 - **The Interaction Effect:** Availability pursuit acts as a success multiplier only when Demand pursuit is high. Strategic search is cognitively wasted if the primary goal is neglected.
 - **The Price Trap:** For solvers with low Availability pursuit, price pursuit acts as a **cognitive distractor** ($\beta = -0.172$). These solvers enter an infinite loop of attempting to purchase unavailable, cheap tickets, failing to adjust their strategy based on system feedback.
- The Optimal Goal Sequence:** Success is maximized through the sequence **Demand > Availability > Price**.

5. Cognitive Synthesis: From Data Points to Behavioral Profiles

The behavioral profiles extracted from the data allow us to differentiate between solvers based on their strategy of conflict resolution. The **Competent Solver** functions as a strategic balancer. They clarify the primary demand and, when met with system failure, systematically explore constraints to find available states, sacrificing price optimality to ensure task completion. In contrast, the **Demand-Locked/Availability-Blind Solver** (Group 10) may understand what they need to buy, but their inability to move beyond a failed state—driven by a blind pursuit of price—renders them ineffective.

Knowledge Representation and Resolution Logic

These patterns reflect the four fundamental components of **Knowledge Representation** in AI logic:

- **Lexical:** Identification of ticket symbols.
- **Structural:** Understanding constraints on symbol usage (Availability rules).
- **Semantic:** Associating the ride requirement with the representation.
- **Procedural:** The goal-balancing strategies used to compute the solution. Critically, the solver's search for available tickets acts as a **Resolution Inference Rule**. In AI logic, resolution proves unsatisfiability by deriving an empty clause through the elimination of complementary literals. Similarly, a proficient solver must "resolve" the conflict between Price and Availability. Their search through ticket attributes is an **Inference Procedure** analogous to reaching a "refutation-complete" proof: they must eliminate the "unavailable" literals (concession tickets) from their knowledge base to arrive at the "resolvent" (a successful purchase). By transitioning from simple metrics like time-on-task to these deep, sequence-based insights, this framework provides a window into the

structure of operative intelligence, moving beyond *if* a problem was solved to *how* the human mind weights priorities and resolves logic-gated conflicts.